

DVIO - Distributed Visual-Inertial Odometry in a Multi-user Environment

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ABSTRACT

Head-mounted displays typically use a visual-inertial odometry system, which relies on the headset's camera combined with Inertial Measurement Units. While effective, this setup fails if the camera is obstructed or if the environments lacks features. Traditional recalibration methods like place recognition often fall short in such settings. Addressing this, the paper proposes a novel distributed tracking method that uses the positions of other users. This approach creates a network or "daisy chain" of user locations, enhancing position tracking accuracy. It serves as an alternative and a supplementary solution to the standard system, ensuring precise location tracking for all users.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Computing methodologies—Artificial intelligence—Computer vision—Computer vision problemsTracking

1 INTRODUCTION

Position tracking in head-mounted displays for mixed reality applications has evolved over the years from inertial odometry, which utilizes an inertial measurement unit (IMU) to determine the 6DoF poses [1], to a hybrid visual-inertial odometry (VIO), which, in addition to the IMU, uses cameras on the headset to localize itself [2]. However, VIO is also prone to errors as it is sensitive to the camera within the head-mounted display (HMD), and once the camera is covered or there is a lack of features in the surroundings, the tracking becomes invalid.

Over the decades, visual tracking systems with external cameras (outside-in tracking) have been thoroughly developed and have high accuracy. However, these systems usually require external devices, resulting in increased cost and reduced flexibility. In comparison, this work proposes a software solution that only relies on the cameras and the positional tracking from the other users' head mounted displays, and is named distributed visual-inertial odometry (DVIO). The proposed system does not generate any additional cost, does not require a previous setup, and is scalable to arbitrary tracking spaces. In this paper, we detail our conceptualization of the proposed system's architecture and functionality using the Microsoft HoloLens 2 (HL2), and present preliminary results.

2 METHOD

The system is based on the concept of distributed localization. Every user inside the system is equipped with an HMD that can track itself, but in addition we use the HMD's camera images to track other users, and this tracking information is shared within the distributed network. In the paper we define "observer" and "observed" for better explanation. An observer is defined as the HMD device which tracks another HMD device, and an observed is defined as the tracked HMD device. In a real world application, an HMD can serve both as

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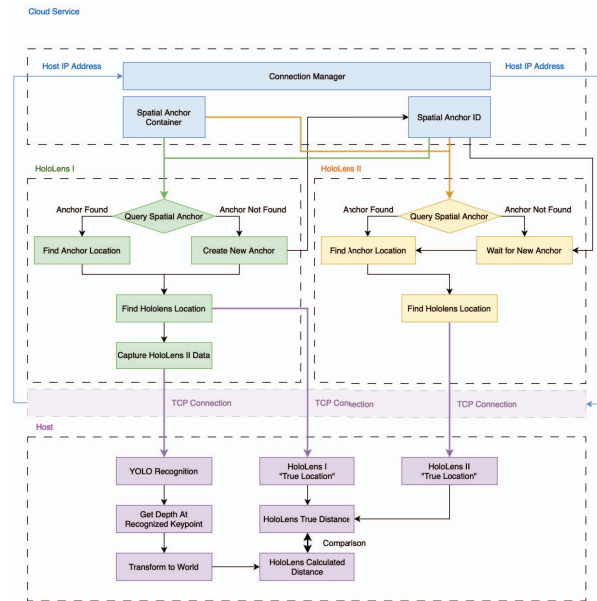


Figure 1: Full application architecture with additional data transmission

observer and observed. The architecture of the proposed solution is shown in Figure 1. The data gathering part consists of collecting 2D images and 3D depth data, along with a corresponding timestamp. The data will be transferred from each HMD to either a cloud server or a local PC. To determine the position of the other HMD, firstly, the body landmarks of the user are extracted from the 2D image. Then, the corresponding 3D world coordinates are obtained using the 3D depth data. The user's spatial transform in the virtual environment is then determined.

2.1 Software and Hardware

Most HMDs have limited computational resources and many of them are allocated to graphical rendering. However, our tracking task also utilizes the GPU, which competes with the graphical rendering task and significantly reduces the frame rate. Therefore, it is better to perform data collection in the HMD as a distributed system, and data processing on a more powerful platform such as a computer or a cloud service as a centralized platform. For data collection in our setup, we use the HL2 since it allows full access to the camera sensor when being operated in research mode. However, the system can be easily migrated to any HMD with 2D image and 3D depth data access.

2.2 Data Collection and Transmission

The data acquisition through the HL2 utilizes Microsoft's research mode API, providing direct access to sensor data instead of processed data in the Windows UWP C++ API. Using this API, the data from the long-throw depth sensor is obtained first. Then, the data coordinates are converted from the camera coordinate to the world

coordinate system using an inverse transform matrix [5]:

$$P^w = M_{f2o}^d (M_{cut}^d)^{-1} P^d \quad (1)$$

where P^w represents the point cloud data in world coordinates; M_{f2o}^d denotes the frame-to-origin matrix; M_{cut}^d is the camera view matrix, and P^d is the raw depth data. When the depth sensor data is captured, the IMU data is simultaneously captured to provide synchronized data, which is important for the accuracy of localization. The API is then encapsulated as a Unity plugin, and functions are called in C# (Unity) to obtain the captured sensor data. The collected sensor data is then processed in Unity and the world coordinates between the HL2 devices are synchronized. The sensor data is sent to the host for data processing via a TCP connection (for further development, a UDP connection can be used for faster data transmission). The data transmission is performed on a local network and can therefore ensure minimal delay and maximal transmission speed. To synchronize the transmission connection, a connection manager is developed on the cloud service.

2.3 Data Processing

The TCP server receives data consisting of the two synchronized datasets - a 2D image and 3D coordinate data as shown in Figure 2, along with the spatial coordinate of the HL2. The original 2D image is adjusted in terms of exposure for better recognition performance and is then fed into a pose tracking model. Then the body landmark detection is performed on the adjusted image, outputting a pose landmark in 2D coordinates from the pixel space. Since the 2D image and 3D data are synchronized in the data collection stage, the corresponding 3D coordinate landmarks of the data can be extracted from the 2D coordinate. The body landmark detection is accomplished with the YOLOv8-pose model, where YOLO (You Only Look Once) is an object detection network that frames the problem as a regression onto spatially separated areas [4], and YOLO-pose is a specified heat-map free approach for joint detection [3].

Then, from the landmark data and the 2D pose from YOLO, the spatial transform of the tracked person is estimated. Additionally, as the data processing system is a centralized platform, it means that the predicted data from the observer can complement the data of the observed, allowing the system to perform a self-correction for more accurate results. In a real life scenario, multiple observers can observe one observed and the localization results can be concatenated. A confidence value, based on the confidence of the observer, occluded landmark, distance between the observer and observed, will be generated. The final data will be calculated as a weighted average of multiple results.

3 PRELIMINARY RESULTS AND NEXT STEPS

The preliminary result of two device tracking is shown in Figure 2. The overall delay of the data processing is around 200 ms and the positional accuracy is within 100 mm. A breakdown of the delay is shown in Figure 3.

The system can be further improved by combining it with existing visual-inertial odometry. For multiple systems to work together, coordinate system synchronization is needed. This synchronization can utilize spatial anchors to perform the task. A spatial anchor is a chosen landmark that remains at the same location in the mixed reality space, regardless of the observing device or time. Once a device observes the spatial anchor, the system will calculate the device's pose relative to the spatial anchor, providing absolute coordinate within the system. In addition to spatial anchors, other systems such as a QR code physical anchor are also viable. To combine distributed tracking with VIO, the confidence levels of the two systems are calculated, and a normalized weighted method is used to estimate the

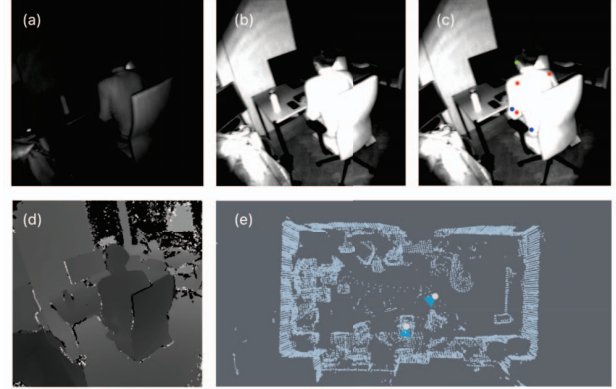


Figure 2: Critical component in data processing and the result: (a) raw IR image from depth sensor, (b) IR image adjusted for tracking, (c) landmark tracking result, (d) 3D coordinate data (e) preliminary result of tracking in a lab environment

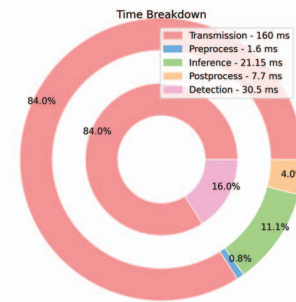


Figure 3: Breakdown of time for one process step

final position of the user. Additionally, to address abrupt displacements resulting from the addition or removal of system components, a vectorized drift during movement is introduced.

4 CONCLUSION

In this paper, DVIO is introduced, which uses distributed tracking to complement the existing visual-inertial odometry. The distributed tracking system consists of a data collection and data processing part, where data collection gathers 2D image and 3D point cloud data from the HL2, and data processing determines a user's location through pose estimation from 2D and 3D data. The next step is combining the distributed tracking system with VIO and evaluate the accuracy and precision of the distributed visual-inertial odometry.

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